

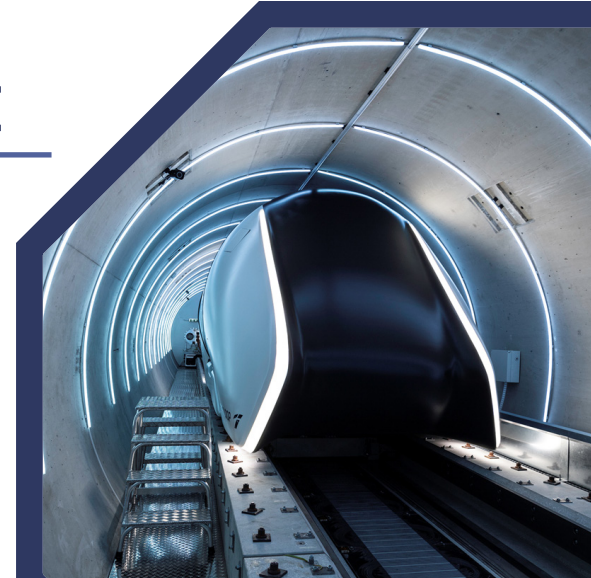
HYPERLOOP | IN REVIEW

DECEMBER 2025

HYPERLOOP'S EVOLVING LANDSCAPE

2025-H2

hyperloop
development
program





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Senator Anjes Tjarks of Hamburg (left) and Domenik Radeck of TUM (right)
at UITP Summit

Dear Hyperloop enthusiasts,

The year 2025 was an important one for Hyperloop in many ways:

Hyperloop has been explicitly mentioned in the **EU General Budget 2025**, following a request from the Council.

The **Hyper4Rail** project, financed by Europe's Rail Joint Undertaking, has kicked off with a very comprehensive consortium that extends beyond the Hyperloop community and is already delivering first results. The project is an ambitious effort to assess technological convergence toward a European standard, define a comprehensive roadmap to Hyperloop's market readiness, conduct a detailed cost-benefit analysis, and validate the technology readiness levels for all elements of Hyperloop. Running until the end of 2026, Hyper4Rail aims to provide answers to key questions the Hyperloop community is facing (see **Chapter 3**).

In November, the **European Commission** published its **Fact-Finding Study** on Hyperloop. The study highlights Hyperloop's potential as an important addition to Europe's transport mix, advancing sustainable mobility and fostering an innovative, competitive industrial sector.

It also confirms the key arguments we have long emphasized: Hyperloop can provide efficient services, reduce travel times, ensure resilience to adverse weather conditions, protect the environment and climate, increase accessibility to remote regions, and much more. The study is somewhat less detailed regarding the cost-benefit analysis, which has since been addressed in the Hyper4Rail project. Nevertheless, it provides a **solid basis for constructive dialogue with the European Commission**, and the overall outcome is more positive than anticipated.

The **Dutch Ministry of Infrastructure and Water Management** has renewed its financial support for HDP, enabling us to expand the team (see photos on the next page) with funding secured through the end of 2026. This provides us with additional capacity to engage more effectively with decision-makers at European, national, and regional levels, as well as with private and institutional investors, in order to secure further support for Hyperloop projects. It will also allow us to enhance our communication efforts, increasing public awareness and understanding of Hyperloop.

The **German Government** has published the **High-Tech Agenda Deutschland**, which explicitly mentions Hyperloop. The responsible ministry has been allocated funding to develop a reference track in Germany. In response, both the City State of Hamburg and the Free State of Bavaria are actively promoting Hyperloop projects.

The **European Hyperloop Center** in Veendam has attracted significant attention as Hardt Hyperloop conducted successful test rides at high speeds, including performing a lane switch maneuver.

EuroTube has opened its mobile facility prototype, a pilot for the fabrication and testing of concrete components at Innovation Park Zurich in Dübendorf, near Zürich.

Zeleros has reached an important milestone by securing a contract with the Turkish government's TÜBİTAK Rail Transportation Technologies Institute (RUTE): a feasibility study for implementing Hyperloop in Turkey, funded by the European Union through Spain's Ministry of Economy, Commerce, and Business.



1) PREFACE (CONT.)

The **Italian HyperTransfer project** has completed its feasibility report and is now preparing for the 'Phase 2' detailed design stage.

The **HDP** has completed the **overhaul of its governance structure** and is entering 2026 with a more transparent and inclusive framework. In addition, it has established its own revenue stream through newly introduced membership fees.

HDP's outreach and stakeholder engagement, together with its members, have culminated in a **series of high-level events and meetings** (see Chapter 2) that have attracted significant attention. The number of institutions and decision-makers involved has grown, and the active participation of Hyperloop promoters has played a key role in the success of these initiatives.

However, important challenges remain. The public and private funding currently available for Hyperloop development is still limited, and further efforts are needed to mobilise additional EU resources. Hyperloop is not yet sufficiently recognised in programs aimed at strengthening European innovation and competitiveness. National governments remain cautious in their support for the technology. As a result, the development of the next larger test infrastructure, as outlined in HDP's Vision Paper, has not progressed as far as it could have.

Sustained support is particularly important to ensure that European Hyperloop promoters, who have driven technological development and positioned Europe as a leader in this field, can continue their work and maintain momentum. Targeted funding, including public contracts and direct investment, would help stabilise growth, support ongoing research and development, and encourage private-sector involvement.

With the help of its recently expanded team, HDP is committed to delivering more positive developments in 2026. Our ambition is to do everything we can to ensure that, by the end of the year, a decision is made and the necessary funding secured to build a larger test site in Europe.

Sincerely,

Klaus Rudischhauser



KLAUS RUDISCHHAUSER



ALGARA CASTLE



CARLOS VILLALBA



MATEO DRAGONI

2) HYPERLOOP DEVELOPMENT IN H2 - 2025

JULY

EHW & HDP Partners Day: The European Hyperloop Week took place in Veendam, with participation from around 50 universities worldwide. During the event, a well-attended HDP Partners Day was held, where the proposal for a new governance and membership structure was presented and discussed. With the new rules in place, HDP will be able to approach more Hyperloop developers and promoters to join as members.

The work program for 2025/26 was introduced and endorsed, with the main discussions focusing on current trends and developments in the Hyperloop ecosystem.

Hamburg Project: A significant consortium for this new and strategically important project is under development and has continued discussions with the authorities of the City State of Hamburg. A first financing request has been submitted to the German Federal Ministry of Economy.

Bavaria Project: The Bavarian government has commissioned a feasibility study to identify potential sites for a reference track.

Policy engagement: HDP engaged with several European Commissioners to highlight the relevance of Hyperloop for European competitiveness, the achievement of EU climate objectives, and regional cohesion. A detailed briefing was also shared with the Director-General of DG RTD, followed by an exchange in October.

The preparation of a series of events scheduled for the second half of the year continued in July, including a Parliamentary Breakfast in Berlin, Hardt's fundraising event in Hamburg, and an event at the Office of the City of Hamburg in Brussels.

Hyper4Rail: An important work package was finalised – a cost-benefit analysis for a Hyperloop network connecting major European airports and cities.

Global developments: HDP partners signed cooperation agreements with partners in India.

AUGUST

Future of HDP: The Dutch government has granted continued funding for 2026, ensuring that HDP can continue to make an impact for another year.

Regional engagement: Contacts were established with the government of the State of Baden-Württemberg and the regional parliament to discuss opportunities for the involvement of high-tech companies in Hyperloop.

S3 Thematic Platform: To facilitate access to innovation funding from DG REGIO, a thematic partnership will be prepared and submitted for registration. This process involves mobilising a number of regions across EU Member States that declare their readiness to join and actively participate.

Rotary Club: HDP gave a presentation on Hyperloop to the Rotary Club of Bad Soden, near Frankfurt.



2) HYPERLOOP DEVELOPMENT IN H2 - 2025 (CONT.)

SEPTEMBER

EuroTube: EuroTube has officially launched its Smart Mobile Factory at the Innovation Park Zurich. The container-based factory is designed for flexible deployment along infrastructure projects at various locations and, in the future, will enable on-site production of concrete components for tube structures.

Investor Event: Hardt organised an investor event in Hamburg, attended by the Senator for Transport and Mobility Transition of the City of Hamburg, Mr. Anjes Tjarks.

Max Bögl: HDP visited the Max Bögl Group, a new member of HDP, and its demonstration facility for the light magnetic levitation train operated by Max Bögl Transport Systems.

OCTOBER

TransPod: The Canadian company TransPod has made significant progress in Alberta, Canada, where the Calgary–Edmonton project has secured strategic partnerships with Algoma Steel and Supreme Steel. The project is now awaiting confirmation from the Government of Alberta of the regulatory framework required to proceed with a full-scale test track as part of a future operational corridor.

Policy engagement: HDP and several partners met with Marc Lemaître, Director-General of DG RTD of the European Commission, to exchange views on the role of Hyperloop in future EU research and innovation funding programmes, including Horizon Europe and the EU's next Multiannual Financial Framework. HDP also held a strategic exchange with DG MOVE at desk-officer level.

An event planned with the City of Hamburg at its Brussels office was postponed and will be rescheduled in 2026.

HDP, together with DRO, met with the Senator for Transport, Environment and Climate Protection of the City of Berlin, Ms Ute Bonde.

HDP, together with several partners, attended the presentation of the High-Tech Agenda Germany with Chancellor Merz and the Minister for Research, Technology and Space, Ms Dorothee Bär. The High-Tech Agenda explicitly mentions plans to construct a reference line for Hyperloop in Germany, and the ministry will make funding available for this purpose.

New HDP Governance Structure: The newly established Steering Committee, composed of representatives of HDP's core partners, met for the first time in Amsterdam. The partners elected their candidates to serve as members of the newly created Supervisory Board.

Project acquisition: The European Hyperloop Center, together with the University of Applied Sciences Emden/Leer and several regional partners, secured an Interreg project (HyperBRIDGE) with financial support of €4.8 million.

The Dutch-German cross-border initiative launched in October 2025, promotes hyperloop innovation through regional collaboration. Supported by the Interreg VI A program, it connects more than 20 SMEs with universities including those operating the European Hyperloop Center and goTube facility (see chapter 4).

Future Perspectives of Hyperloop: ARUP and Hyper4Rail held a seminar in Amsterdam to discuss current trends in Hyperloop development. The event featured in-depth discussions on both the short-term and long-term perspectives of the technology, and the cost-benefit analysis produced by Hyper4Rail was presented and reviewed.

2) HYPERLOOP DEVELOPMENT IN H2 - 2025 (CONT.)

NOVEMBER

Policy Dialogue: HDP, together with several partners, hosted a well-attended Parliamentary Breakfast for members of the German Parliament. HDP also met with a representative of the European Investment Bank to explore the organisation of a presentation on Hyperloop with one of the bank's Vice-Presidents in 2026.

European Commission: The EU Commission published the long-awaited fact-finding study on Hyperloop, which provides a comprehensive assessment of its potential. HDP welcomed the study's findings and submitted a response to the Commissioner in charge and the relevant Commission services, highlighting opportunities to build on these insights and to advance European Hyperloop development through targeted funding and long-term strategic support, while also noting the Commission's commitment to establishing a European Hyperloop Action Plan (see Chapter 6).

The European Commission also published a communication on High-Speed Rail for 2040.

The document outlines a roadmap and proposed measures aimed at developing a more efficient and faster high-speed rail network across Europe. It includes the goal of significantly reducing travel times on many key rail routes, which raises important considerations for the Hyperloop ecosystem and its future integration with European transport networks.

Global developments: HDP partners and Hyper4Rail attended the 5th International Workshop on Evacuated Tube Maglev Transport 2025 in Chengdu, China which included a presentation on the progress of Southwest Jiaotong University's hyperloop test site for high-temperature superconducting technology. Swisspod officially inaugurated the almost completed new test site in Pueblo, Colorado and unveiled its new pod Aerys 1.

Hyperloop Conference: The 2025 Hyperloop Conference was held during the Smart City Summit in Barcelona, with the Hyper4Rail consortium meeting back-to-back with the conference to review and discuss the project's progress.

DECEMBER

New HDP Team Members: With continued financial support from the Dutch government, HDP welcomed three new team members: Carlos Villalba for project development, Algara Castle for public affairs and communications, and Matteo Dragoni for research infrastructures. Their recruitment strengthens HDP's capacity to advance projects, engage with stakeholders, and develop research infrastructures.

Outreach and visibility: A comprehensive program of events for 2026 was prepared, and implementation began immediately. A flagship initiative will be a joint HDP stand with partners at INNOTRANS 2026 in September.

China and India: HDP partners produced summary papers on Hyperloop activities in China and India, which will be made available to all HDP Partners and used in outreach activities..



3) HYPER4RAIL – PROGRESS AND RESULTS ACHIEVED IN 2025

In its first year, the EU-funded Hyper4Rail project has delivered substantial progress. Bringing together public and private partners to develop a harmonised European hyperloop concept and establish a clear roadmap for the future development and deployment of hyperloop in Europe, work over the past 12 months focused on **operations concepts and the cost–benefit analysis of a European hyperloop network**. Key achievements include:

A comprehensive review of European transport policy objectives, stakeholder needs, and system-level challenges, combined with an overview of existing and emerging transport technologies and hyperloop concepts. This work establishes a common technical and strategic foundation for future harmonisation (*D2.1: Hyperloop Technologies, Applications & Opportunities*).

A quantitative cost–benefit analysis, aligned with EU guidelines, confirms that hyperloop can be economically viable for long-distance European travel. An integrated rail-hyperloop network connecting all European capitals at >750 km/h could shift nearly 50% of intra-European air traffic, with financial break-even at around €115 million/km (*D3.1: Hyperloop Network Analysis and Use Case Identification*).

A first set of system requirements and technology comparisons was completed, covering vehicle size, aerodynamics, tube loads, levitation, propulsion, cooling, and control systems.

Stakeholder consultations confirmed that hyperloop's highest added value lies on long-distance corridors (>600 km), primarily by shifting demand from aviation. Integration with rail and other modes, interoperability, and end-to-end travel time reduction emerged as core priorities.

A preliminary concept of operations was defined, highlighting the importance of very high speeds, direct connections, fast switches, and short boarding times. A common system architecture and initial safety targets were agreed among developers, inspired by EU rail and aviation safety frameworks (*D2.2: Interim Hyperloop Design Concept and Business Case*).

Finally, a first corridor-level business case was developed for a France–Italy connection, demonstrating the applicability of the approach to concrete European routes.

HYPER4RAIL PROGRESS AND RESULTS ACHIEVED IN 2025



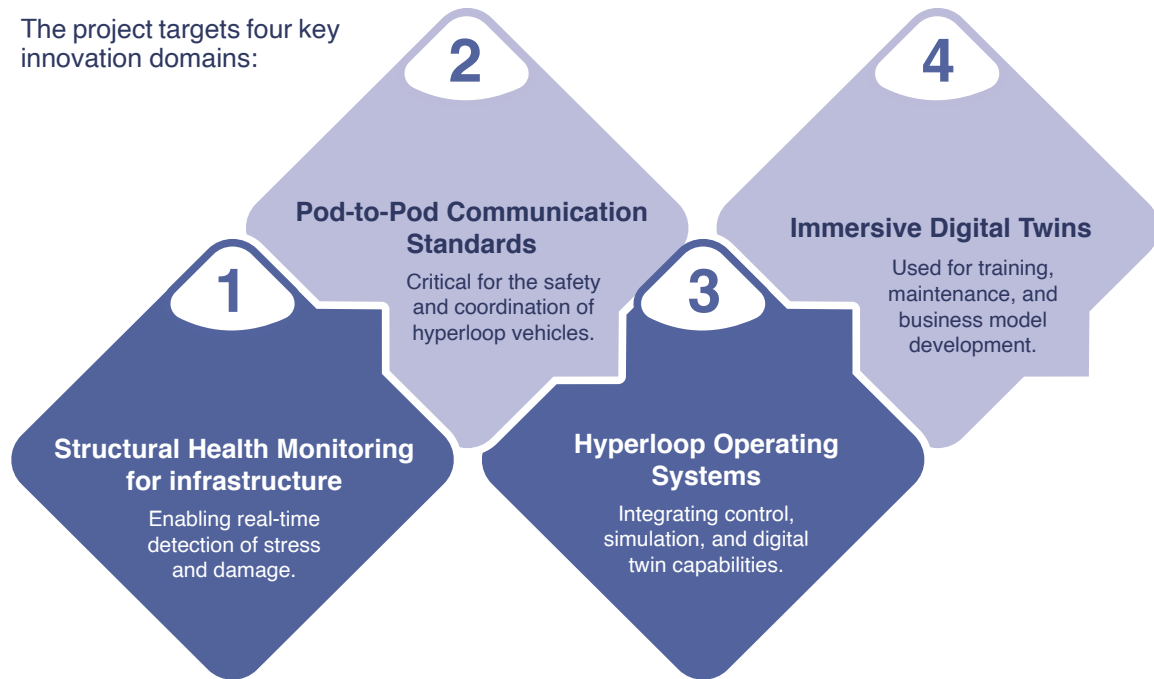
4) HYPERBRIDGE: DUTCH-GERMAN CROSS-BORDER HYPERLOOP INNOVATION PROJECT

The HyperBRIDGE project was launched in 2025 with support from the Interreg VI A Germany–Netherlands programme. This €4.8 million cross-border initiative brings together businesses, universities, and research institutes to position the region as a European frontrunner in the development of hyperloop technologies.

The project aims to strengthen the regional innovation ecosystem by connecting at least 20 SMEs with major knowledge institutions through shared research, testing, and training programs. The project builds on the assets of the European Hyperloop Center (EHC) in Veendam and the goTube test facility at the Hochschule Emden/Leer.

Beyond hyperloop-specific applications, HyperBRIDGE aims to generate spill-over effects for related sectors including logistics, automotive, energy, and infrastructure.

The project targets four key innovation domains:



In addition, the project will develop a serious game to simulate new business models and a Digital Knowledge Hub to facilitate long-term collaboration and open innovation. The HyperBRIDGE consortium consists of companies and universities, including Omnidots, Concept7, V-Tron and Quantumfrog, alongside leading universities like Rijksuniversiteit Groningen, Hanzehogeschool Groningen, and Hochschule Emden/Leer. The initiative is coordinated by the European Hyperloop Center.

With expectations of six companies committing to follow-up projects and anticipated European market value reaching €10 billion annually, HyperBRIDGE positions the region as a leading hyperloop innovation hub.

h 5) HYPERLOOP TEST FACILITIES IN 2025

UNDER CONSTRUCTION

- 10. Chengdu test track (CN)
- 11. DemoTube (CH)
- 12. Transpod test track in Limoges (France)



OPERATIONAL FACILITIES

- 1. European Hyperloop Center (NL)
- 2. goTube test facility (DE)
- 3. ITT Madras test track (IN)
- 4. TUM Hyperloop test track (DE)
- 5. Datong Shanxi test track (CN)
- 6. Nevomo test track (PL)
- 7. Swisspod test track (CH)
- 8. Tubitak (TR)
- 9. Swisspod test track (USA)



PLANNED

- 13. Veneto (IT)
- 14. Ciudad Real (SP)
- 15. Reference Track in Hamburg
- 16. Reference Track in Bavaria

A large white cylindrical vessel, likely a test chamber, is being lifted by a white crane with a green boom. The crane has "www.consensa.ch" written on it. The vessel is suspended by red slings. In the background, there is a white rectangular structure with a circular opening. A worker in a yellow safety vest and white hard hat is standing near the vessel. The scene is outdoors on a paved area with a blue sky and clouds.

Construction of DemoTube test facility in Dübendorf, Switzerland

DEVELOPMENTS AT EU LEVEL



The EU Commission has published the results of the Fact-Finding Study on Hyperloop carried out by Ramboll, TIS, CERN and Sintef. The study adopts an overall positive tone, noting that while Hyperloop is not yet ready for commercial deployment, it is maturing rapidly and could soon progress from prototyping to demonstration. It highlights Hyperloop's potential to support decarbonisation, digitalisation, regional cohesion, and industrial renewal, identifying the technology as strategically relevant for the EU and therefore recommending that the Commission take further action on a range of issues.

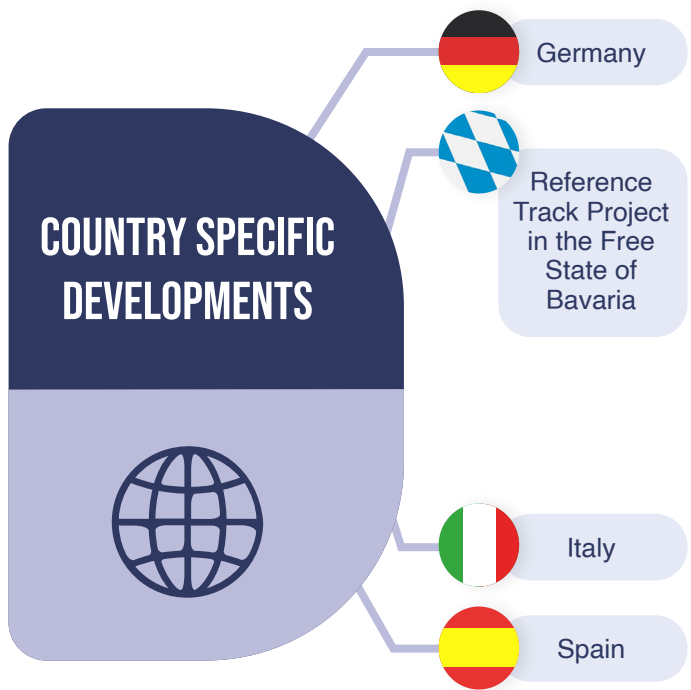
However, the study is lacking some of the analytical elements requested in the Tender Specifications for the study. In particular, it does not provide a quantitative and monetary assessment of economic, social, and environmental impacts, nor does it make systematic use of analytical tools such as cost-benefit analysis, sensitivity analysis, econometric modelling, or risk analysis to address the gap for a clear and unbiased business case for hyperloop. The main reason given for these limitations is the current lack of sufficiently reliable data.

Apostolos Tzitzikostas, EU Commissioner for Transport and Mobility, has not yet put forward a plan of action, but this is expected following the publication of the study. HDP has offered its assistance to the Commission and submitted its response to the study before the end of the year.

The European Commission has also published a communication on Connecting Europe through High-Speed Rail, which aims to complete the TEN-T network. The communication sets out an ambitious vision, but its implementation raises questions in light of the budgetary constraints faced by many Member States. In this context, there is scope for Hyperloop to position itself as a complementary solution. HDP is currently preparing a position paper on this matter.

Hyperloop is not yet fully reflected in key European policy frameworks related to competitiveness, sustainability, and research and innovation. The ongoing negotiations on the next Multiannual Financial Framework therefore represent an important opportunity to further integrate Hyperloop into these strategic priorities.

Within the European Parliament, the lead Committee on Industry, Research and Energy (ITRE) has appointed rapporteurs for two key dossiers. EPP MEP Dr Christian Ehler will serve as rapporteur for the Horizon Europe Framework Programme for Research and Innovation 2028–2034 (the future FP10) and, together with the ITRE Coordinator of the S&D Group, Ines Galvez, as co-rapporteur for the European Competitiveness Fund. HDP will reach out to the rapporteurs to engage in discussions on the relevance of Hyperloop in the context of these forthcoming initiatives.



Concrete plans are being advanced by the City State of Hamburg and the Free State of Bavaria, both of which are proposing the construction of a first reference track. At the federal level, the German Ministry for Research, Technology and Space has received a mandate and dedicated funding to support the development of a Hyperloop reference line. However, a more precise implementation framework has not yet been defined.

The Free State of Bavaria has continuously supported the development of Hyperloop technology since 2020 through a dedicated research programme at the Technical University of Munich (TUM) under the High-Tech Agenda Bavaria and has recently confirmed the continuation of this support in the coming years. Following the completion of the full-scale Hyperloop demonstrator in 2023, the state additionally funded a feasibility study in 2025 to prepare the implementation of a reference track in Bavaria.

The study, recently completed by Neoways and TUM together with two planning offices, identified and assessed several potential sites. Based on these results, the Bavarian State Government has reaffirmed its commitment to site acquisition and to advancing the realisation of the project in Bavaria.

In parallel, Neoways and TUM have built a strong international network of leading industry and research partners in preparation for the project kick-off.

The consortium for the Veneto project has been restructured. Beyond this, no further developments have come to HDP's attention during the period under review.

The Spanish government has indicated that a Hyperloop project in Castilla-La Mancha is currently not among the government's immediate priorities. Discussions did not advance further during the reporting period.

Zeleros has reached an important milestone by securing a contract with the Turkish government's TÜBİTAK Rail Transportation Technologies Institute (RUTE): a feasibility study for implementing Hyperloop in Turkey, funded by the European Union through Spain's Ministry of Economy, Commerce, and Business.



The Dutch government continues to provide funding for HDP and the European Hyperloop Center. This sustained commitment is essential in maintaining Hyperloop activities in Veendam in the current funding environment.

The government supports TÜBİTAK RUTE, and a test facility has been built. Hyperloop-related activities in Turkey are progressing at a steady pace.

With strong support from the government, Hyperloop-related activities are progressing steadily, as reflected in the country paper on India recently published by HDP.

Significant resources are being invested in Hyperloop development at several competing locations in China. These developments are outlined in the country paper recently published by HDP.

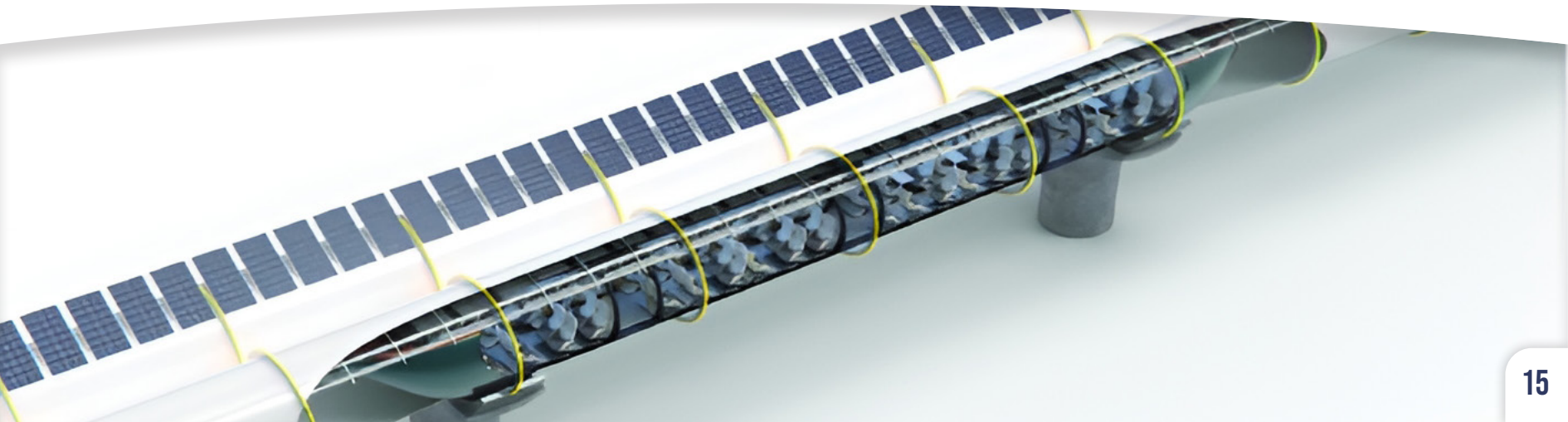
The Swisspod facility in Colorado is nearing completion. It will be the biggest test site so far. Initial test runs have already been carried out successfully at very high speeds with a new pod, Aerys 1.

Preparatory work on a large-scale project connecting Calgary and Edmonton is progressing. The project is now awaiting confirmation from the Government of Alberta of the regulatory framework required to proceed with a full-scale test track as part of a future operational corridor.



7) FUTURE OUTLOOK

The main challenge for the European Hyperloop ecosystem is to remain at the forefront of global developments. To date, Europe can credibly claim a leading position in Hyperloop technology:





7) FUTURE OUTLOOK (CONT.)

European promoters have achieved several important milestones in the development of Hyperloop technology, including certified prototype vehicles and significant test infrastructure developments. These achievements demonstrate Europe's strong technological position, but this potential still needs to be fully reflected in public policy frameworks. Ongoing initiatives at European, national, and regional levels show that this alignment is beginning to take shape. In particular, developments in Bavaria and Hamburg position Germany as a key driver of European Hyperloop, with the new year expected to provide clearer indications of tangible progress.

At the same time, European Hyperloop developers continue to face financial constraints as the sector transitions from research and prototyping toward demonstration and deployment. Private investment remains limited, and public funding mechanisms are still evolving. HDP has therefore intensified its dialogue with public authorities at regional, national, and European levels to support the mobilisation of public resources. The recent strengthening of the HDP team is intended to enhance its capacity to address these challenges and to support the ecosystem more effectively.

Work is underway to align the activities of the European Hyperloop Center, the European Hyperloop Week, and HDP in order to strengthen the overall visibility and coherence of Hyperloop initiatives in Europe. The new governance structure provides a framework to support the active engagement of new partners, contributing to increased representativeness and impact in 2026.

The Hyper4Rail project, with its ambitious objectives, plays a central role in shaping the future of Hyperloop in Europe. Its results are expected to inform policymaking and contribute to the development of supportive frameworks at both European and national levels. Encouragingly, after a period of predominantly sceptical media coverage, public reporting on Hyperloop in 2025 shifted toward a more positive tone, increasingly focusing on concrete developments and the technology's relevance. Taken together, these developments provide a solid basis for cautious optimism as Europe enters a decisive phase for Hyperloop in 2026.



**goTube test facility
in Emden, Germany**





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2025-H2

HYPERLOOP IN REVIEW